

Assignment 1

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1 Poisson-gamma model

The dataset contains $N = 500$ observations, with mean $\bar{x} \approx 5.34$, median equal to 5 and a range of values from 0 to 17

We assume:

$$x_i = \text{Poisson}(\lambda) \quad (1)$$

$$\lambda = \text{Gamma}(\alpha, \beta) \quad (2)$$

with prior hyperparameters (in absence of knowledge) $\alpha = 1, \beta = 1$.

1.1 Posterior Distribution

The posterior distribution is:

$$p(\lambda \mid D) = \text{Gamma}(\alpha + \sum x_i, \beta + N)$$

So the hyperparameters are $\alpha_{\text{post}} = \alpha + \sum x_i$ and $\beta_{\text{post}} = \beta + N$. The MAP estimate for a Gamma distribution is:

$$\lambda_{\text{MAP}} = \frac{\alpha_{\text{post}} - 1}{\beta_{\text{post}}} \approx 5.33$$

The posterior mean and variance are:

$$\mathbb{E}[\lambda] = \frac{\alpha_{\text{post}}}{\beta_{\text{post}}} \quad (3)$$

$$\text{Var}(\lambda) = \frac{\alpha_{\text{post}}}{\beta_{\text{post}}^2} \quad (4)$$

Giving as a result:

$$\lambda \approx 5.33 \pm 0.10$$

1.2 Discussion

Since α_{post} is very large, the Gamma posterior is approximately symmetric and a Gaussian distribution is a good approximation. The error is small due to the large sample size.

2 Posterior Predictive Distribution

In order to derive the distribution of a new observation x_{new} given the observed data D ($p(x_{\text{new}} \mid D)$). Since λ is unknown, we marginalize it out using the posterior distribution:

$$p(x_{\text{new}} \mid D) = \int p(x_{\text{new}} \mid \lambda) p(\lambda \mid D) d\lambda$$

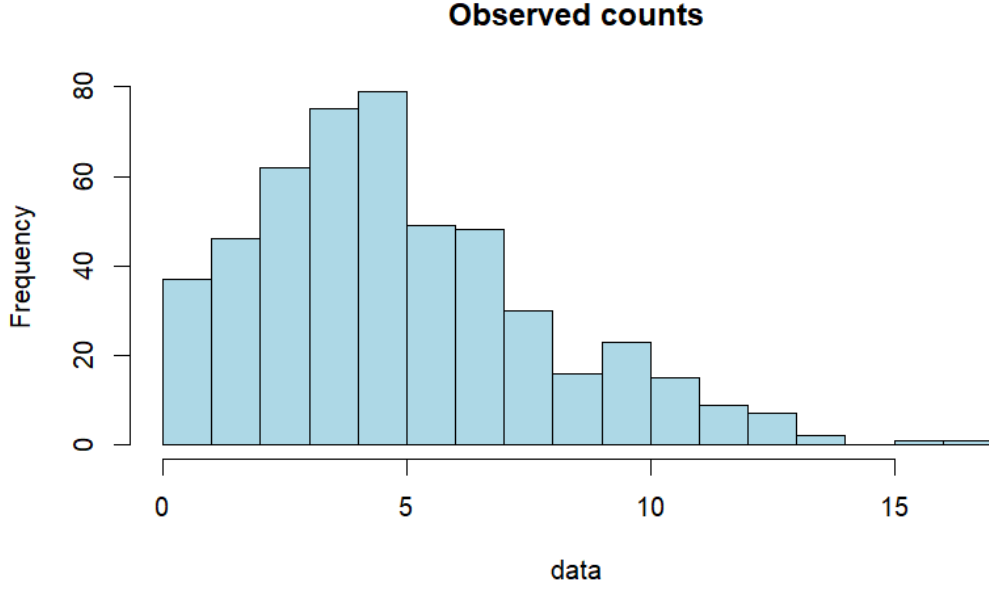


Figure 1: Data frequency

The likelihood for a new observation is given by:

$$p(x_{\text{new}} | \lambda) = \frac{\lambda^{x_{\text{new}}} e^{-\lambda}}{x_{\text{new}}!}$$

Given that $\lambda | D \sim \text{Gamma}(\alpha_{\text{post}}, \beta_{\text{post}})$, the density is given by:

$$p(\lambda | D) = \frac{\beta_{\text{post}}^{\alpha_{\text{post}}}}{\Gamma(\alpha_{\text{post}})} \lambda^{\alpha_{\text{post}}-1} e^{-\beta_{\text{post}} \lambda}$$

Combining the two expressions:

$$p(x_{\text{new}} | D) = \int \frac{\lambda^{x_{\text{new}}} e^{-\lambda}}{x_{\text{new}}!} \cdot \frac{\beta_{\text{post}}^{\alpha_{\text{post}}}}{\Gamma(\alpha_{\text{post}})} \lambda^{\alpha_{\text{post}}-1} e^{-\beta_{\text{post}} \lambda} d\lambda \quad (5)$$

$$= \frac{\beta_{\text{post}}^{\alpha_{\text{post}}}}{x_{\text{new}}! \Gamma(\alpha_{\text{post}})} \int \lambda^{x_{\text{new}} + \alpha_{\text{post}} - 1} e^{-(\beta_{\text{post}} + 1)\lambda} d\lambda \quad (6)$$

The integral can be seen as the Gamma distribution:

$$\int \lambda^{a-1} e^{-b\lambda} d\lambda = \frac{\Gamma(a)}{b^a}$$

Thus follows the new expression:

$$p(x_{\text{new}} | D) = \frac{\Gamma(\alpha_{\text{post}} + x_{\text{new}})}{x_{\text{new}}! \Gamma(\alpha_{\text{post}})} \left(\frac{\beta_{\text{post}}}{\beta_{\text{post}} + 1} \right)^{\alpha_{\text{post}}} \left(\frac{1}{\beta_{\text{post}} + 1} \right)^{x_{\text{new}}}$$

The final result is that the new observation corresponds to a Negative Binomial distribution:

$$x_{\text{new}} \sim \text{NegBin} \left(\alpha_{\text{post}}, \frac{\beta_{\text{post}}}{\beta_{\text{post}} + 1} \right)$$

2.1 Probability test

We compute:

$$P(x_{\text{new}} > 2\bar{x})$$

Using the empirical mean:

$$2\bar{x} \approx 10.68$$

Numerically:

$$P(x_{\text{new}} > 2\bar{x}) \approx 0.021$$

This probability is small, indicating that large deviations from the mean are rare but still possible.

3 Negative Binomial Model

3.1 Posterior Distribution and Choice of Priors

We consider the model:

$$x_i \sim \text{NegBin}(r, \pi)$$

where $r > 0$ is a dispersion parameter and $\pi \in (0, 1)$ is a probability parameter. Assuming independence, the likelihood is:

$$p(D | r, \pi) = \prod_{i=1}^N \binom{x_i + r - 1}{x_i} \pi^r (1 - \pi)^{x_i}$$

The choice of prior distributions must respect the domain of the parameters and should ideally be weakly informative in the absence of strong prior knowledge. Since $r > 0$, a natural choice is a Gamma distribution: it is flexible and defined on the positive real line. Since $\pi \in (0, 1)$, we choose a Beta distribution. In the absence of prior information, we adopt $r \sim \text{Gamma}(1, 1)$ and $\pi \sim \text{Beta}(1, 1)$: an exponential distribution with mean 1 and a uniform distribution on $(0, 1)$

3.2 Posterior Distribution

Using Bayes' theorem:

$$p(r, \pi | D) \propto p(D | r, \pi) p(r) p(\pi)$$

Substituting the likelihood and priors:

$$p(r, \pi | D) \propto \left[\prod_{i=1}^N \binom{x_i + r - 1}{x_i} \pi^r (1 - \pi)^{x_i} \right] \cdot r^{a-1} e^{-br} \cdot \pi^{c-1} (1 - \pi)^{d-1} \quad (7)$$

Rearranging terms:

$$p(r, \pi | D) \propto \left[\prod_{i=1}^N \binom{x_i + r - 1}{x_i} \right] r^{a-1} e^{-br} \cdot \pi^{Nr+c-1} \cdot (1 - \pi)^{\sum x_i + d - 1} \quad (8)$$

3.3 MAP Estimation and Uncertainty

Using numerical optimization, we obtain:

$$r_{\text{MAP}} \approx 6.51 \pm 0.87 \quad (9)$$

$$\pi_{\text{MAP}} \approx 0.549 \pm 0.034 \quad (10)$$

We obtain that π is well estimated (small uncertainty) and r has larger uncertainty

3.4 Model Evidence and Bayes Factor

In Bayesian model comparison, the evidence is defined as:

$$p(D | H) = \int p(D | \theta) p(\theta) d\theta$$

where θ denotes the model parameters.

For the Negative Binomial model, $\theta = (r, \pi)$, hence:

$$p(D | H_{\text{NB}}) = \int p(D | r, \pi) p(r, \pi) dr d\pi$$

This integral cannot be computed analytically but it is possible to approximate by using the Laplace approximation, which is justified when the posterior is sharply peaked around its maximum.

Let $(r_{\text{MAP}}, \pi_{\text{MAP}})$ be the maximum a posteriori estimate, and let Σ be the covariance matrix obtained from the inverse Hessian of the negative log-posterior.

Then:

$$p(D | H_{\text{NB}}) \approx p(D | \hat{\theta}) p(\hat{\theta}) (2\pi)^{k/2} |\Sigma|^{1/2}$$

Taking the logarithm:

$$\log p(D | H_{\text{NB}}) \approx \log p(D | \hat{\theta}) + \log p(\hat{\theta}) + \frac{k}{2} \log(2\pi) + \frac{1}{2} \log |\Sigma|$$

where $k = 2$ is the number of parameters.

Using the MAP estimate and the Hessian computed numerically, we obtain:

$$\log p(D | H_{\text{NB}}) \approx -1242.21$$

For the Poisson model with Gamma prior, the evidence can be computed analytically.

From:

$$p(D | \lambda) \propto \lambda^{\sum x_i} e^{-N\lambda}, \quad \lambda \sim \text{Gamma}(\alpha, \beta)$$

the marginal likelihood is:

$$p(D | H_{\text{Poisson}}) = \frac{\beta^\alpha}{\Gamma(\alpha)} \frac{\Gamma(\alpha + \sum x_i)}{(\beta + N)^{\alpha + \sum x_i}} \prod_{i=1}^N \frac{1}{x_i!}$$

Taking the logarithm:

$$\log p(D | H_{\text{Poisson}}) \approx -1280.18$$

The Bayes Factor comparing the two models is:

$$BF = \frac{p(D | H_{\text{NB}})}{p(D | H_{\text{Poisson}})}$$

or in logarithmic form:

$$\log BF = \log p(D | H_{\text{NB}}) - \log p(D | H_{\text{Poisson}})$$

Substituting the numerical values:

$$\log BF \approx -1242.21 - (-1280.18) = 37.97$$

$$BF = e^{37.97} \approx 3.1 \times 10^{16}$$

The Bayes Factor is $BF \gg 1$, indicating a great evidence in favor of the Negative Binomial model, also reflecting the fact that the Poisson model imposes the constraint $\text{Var}(X) = \mathbb{E}[X]$ while the Negative Binomial model allows $\text{Var}(X) > \mathbb{E}[X]$

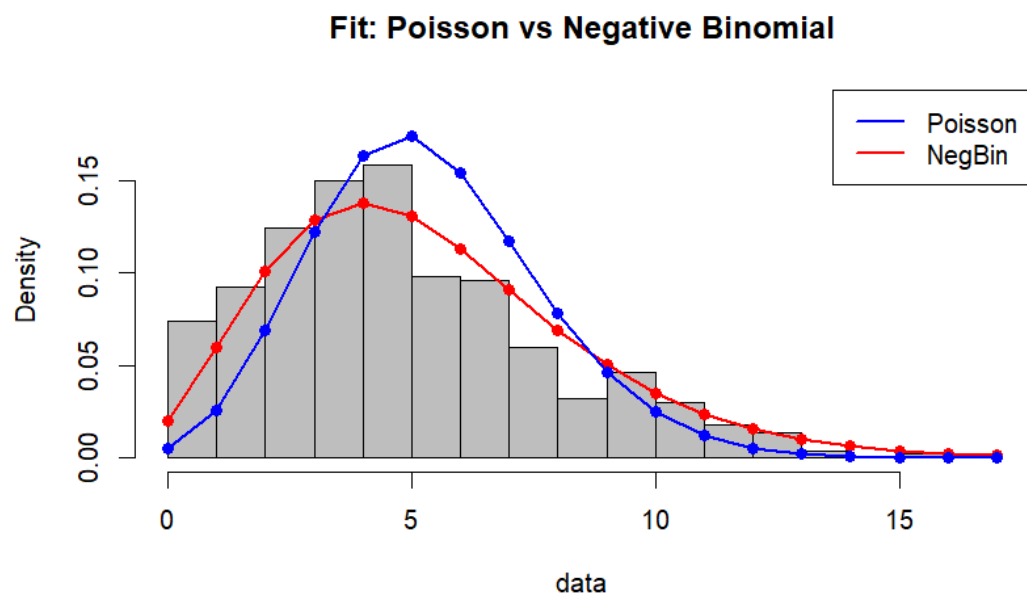


Figure 2: Posterior comparision